SMALL MODULI PROBLEMS OVER $\operatorname{Spec}\mathbb{Z}$: QUADRATIC ALGEBRAS, ROTATIONS IN PGL_2 AND THE INTEGRAL SCHWARZIAN, AFTER GROTHENDIECK'S WORKING NOTES

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Abstract. Five folders of Grothendieck's notes from the 1980s, kept at the University of Montpellier, treat small moduli problems over $\operatorname{Spec}\mathbb{Z}$ without inverting 2, 3 or 5. We prove the parts that can be done in coordinates over an arbitrary commutative ring, and correct the pages where they are wrong. Quadratic algebras are classified by triples (L, δ, T_0) when 2 is a non-zero-divisor; their product has discriminant $\delta_1\delta_2$ and is the invariant algebra of $\sigma_1 \otimes \sigma_2$ after every base change exactly when 2, b_1, b_2 generate the unit ideal. The Hamilton–Cayley structures on $H \otimes H'$ form an affine line, and a structure is Azumaya exactly when $Q(t) = t^2 - bb't + b^2c' + b'^2c - 4cc'$ is a unit, in characteristic 2 as well; Q is the equation of the product of H and H'. The invariant $A(g) = (\operatorname{Tr} g)^2/\det g - 2$ of $g \in \mathrm{PGL}_2$ detects the orders 3 to 6; we prove rigidity and a normal form for the group G_2, and show that in characteristic 5 the octahedral configuration fills $\mathbb{P}^1(\mathbb{F}_5)$. The Schwarzian written with divided derivatives is a cocycle over $\mathbb{Z}$, and its vanishing forces a homography at Frobenius level p for $p > 3$ and p^2 for $p = 2, 3$. The proofs are formalised in Lean.

1. Introduction

Five folders of Grothendieck's working notes, kept at the University of Montpellier, treat small moduli problems over an arbitrary base:

- folder 84, on binary quadratic forms and quadratic algebras, dated by the inventory from 1982 to about 1986;
- folder 83, on Azumaya algebras of rank 4, with a part dated April 1986;
- folder 85, *2-polyèdres et 1-polygônes*, on regular polygons and polyhedra realised in forms of PGL_2;
- folder 64, *Modules courbes elliptiques en caractéristiques* $\neq 2$, dated from 1980–1982;
- folder 93, on the Schwarzian derivative and projective connections.

The method is the same throughout. Grothendieck writes the objects in coordinates over $\mathbb{Z}$, finds their invariants, and looks at what happens at the primes 2, 3 and 5 that the classical theory inverts. This paper states and proves the results of these folders that can be established in elementary terms. A statement is attributed to Grothendieck, with the page, when the page states it; otherwise it is said to be *after* the page, and we say what was changed. The proofs are ours unless we say that we follow the page. Every proof given here is formalised in Lean 4 with Mathlib (Section 6), except where we say otherwise. Grothendieck's global moduli statements need more than coordinates; we quote them as statements and say what we prove of them.

Throughout, R is a commutative ring.

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2. Quadratic algebras and their product

Folder 84 works with the quadratic algebras $A(b,c) = R[U]/(U^2+bU+c)$, without dividing by 2, and their discriminant $\delta(b,c) = b^2 - 4c$. With this sign, $\mathrm{Tr}(U) = -b$ and the canonical involution is $\sigma(U) = -b - U$.

Over a general affine base, page 45 describes a quadratic cover by an invertible module L (for $A(b,c)$, L is free, generated by the class of U in A/R), an element $b \in L$ and an element $c \in L \otimes L$. The *triple* of the cover is (L, δ, T_0), where $\delta = b \otimes b - 4c$ and T_0 is the class of b in $L/2L$. A morphism of covers $(L,b,c) \to (L',b',c')$ is a pair (φ, t), where $\varphi\colon L \to L'$ is an isomorphism, $t \in L'$, $\varphi(b) = b' + 2t$ and $(\varphi\otimes\varphi)(c) = c' + b' \otimes t + t \otimes t$. A morphism of triples is an isomorphism φ with $(\varphi\otimes\varphi)(\delta) = \delta'$ and $\varphi(T_0) = T_0'$ in $L'/2L'$.

Theorem 2.1 (Grothendieck, folder 84, pp. 45–53, affine base)**.** *Let L, L' be invertible R-modules.*

1. *Over every R, the triple of a cover satisfies $T_0^2 \equiv \delta \bmod 4$, that is, $\tau\otimes\tau - \delta \in 4(L\otimes L)$ for a lift τ of T_0, and this does not depend on the lift. A morphism of covers is a morphism of triples, and every triple with $T_0^2 \equiv \delta \bmod 4$ is the triple of a cover with the same L.*
2. *If 2 is a non-zero-divisor in R, every morphism of triples lifts to a unique morphism of covers. The functor from covers to triples is then an equivalence of groupoids. It commutes with base change $R \to S$ on objects and morphisms, over every R.*
3. *Over $\mathbb{Z}$, an integer δ is a discriminant iff $\delta \equiv 0, 1 \bmod 4$, and $A(b,c) \cong A(b',c')$ iff $\delta(b,c) = \delta(b',c')$ (p. 53).*

Sketch. One has $b\otimes b - \delta = 4c$, and $(\tau - 2a)\otimes(\tau-2a) - \tau\otimes\tau = 4(a\otimes a - \tau\otimes a)$, using that $L\otimes L$ is symmetric for L invertible. If $\tau\otimes\tau - \delta = 4k$, the cover (L,τ,k) has triple (L,δ,T_0): this is the formula $c = (\tau^2-\delta)/4$ of the page, written without division. When 2 is regular, it is regular on the projective modules L' and $L'\otimes L'$. The equation $\varphi(b) = b' + 2t$ then determines t, and the second condition on (φ,t) follows from $(\varphi\otimes\varphi)(\delta) = \delta'$ after cancelling 4. Over $\mathbb{Z}$, $(b'-b)(b'+b) = 4(c'-c)$ forces $b' \equiv b \bmod 2$ once the discriminants agree. □

Remark 2.2. The hypothesis on 2 in (2) is needed. Over $\mathbb{F}_2$, $A(1,0) = \mathbb{F}_2\times\mathbb{F}_2$ and $A(1,1) = \mathbb{F}_4$ have the same b and the same $\delta = 1$, and they are not isomorphic, since a change of variable $V = U + t$ sends $(1,c)$ to $(1, c+t+t^2) = (1,c)$. Pages 49–52 reconstruct the algebra, its trace and its norm from the triple without division. In the free case, an element $y = a + eU$ has coordinates $\beta = \mathrm{Tr}\, y$ and $\gamma = e$, with $\beta \equiv \gamma b \bmod 2$, so that $y = (\beta + \gamma z)/2$ with $z = 2U + b$ and $z^2 = \delta$. The unit is $(2,0)$, and

$$2\beta(yy') = \beta\beta' + \gamma\gamma'\delta, \qquad 2\gamma(yy') = \beta\gamma' + \gamma\beta', \qquad 4N(y) = \beta^2 - \gamma^2\delta.$$

For non-trivial L we prove only that each of these divisions is legitimate; we do not prove that the pairs (β,γ) form a ring isomorphic to the algebra of the cover.

The product. Page 63 defines the product of two quadratic algebras, "sauf erreur", by $b = b_1b_2$ and $4c = b^2 - \delta_1\delta_2$. The boxed formula writes $4c_1c_1$ where the computation gives $4c_1c_2$. Page 64 checks the generator.

Proposition 2.3 (Grothendieck, folder 84, pp. 63, 64, 89, 102)**.** *Let u_i satisfy $u_i^2 + b_iu_i + c_i = 0$ in commutative R-algebras A_i, and put $b = b_1b_2$ and $c = c_1b_2^2 + c_2b_1^2 - 4c_1c_2$. In any commutative ring, with $Q_i = u_i^2 + b_iu_i + c_i$ and $u = 2u_1u_2 + b_1u_2 + b_2u_1$,*

$$u^2 + bu + c = b_2^2Q_1 + b_1^2Q_2 + 4(Q_1Q_2 - c_1Q_2 - c_2Q_1).$$

Hence $\mathcal{U} = 2U_1U_2 + b_1U_2 + b_2U_1 \in A_1 \otimes_R A_2$ satisfies $\mathcal{U}^2 + b\mathcal{U} + c = 0$ and defines a morphism of R-algebras $A(b,c) \to A_1\otimes A_2$. Moreover $b^2 - 4c = \delta_1\delta_2$.

Proof. Expand both sides; the identity is polynomial in u_1, u_2, b_i, c_i, with integer coefficients. The universal property of $A(b,c)$ gives the morphism. □

Voight studies this product over an arbitrary base scheme as the monoid of quadratic rings [5]. His Construction 3.12 is the same formula in the opposite sign convention; he credits the free affine case to Hahn and proves that the monoid structure is unique (Theorem 3.24). Part (iii) of that theorem identifies the product with the invariants of $\sigma_1 \otimes \sigma_2$ when both algebras are separable. For quadratic algebras over rings in general, see Knus [4]. Pages 84–86 prove the identification with the invariants under a weaker hypothesis.

Theorem 2.4 (Grothendieck, folder 84, pp. 84–86)**.** *Let $A_i = A(b_i, c_i)$, with bases $(1, U_i)$, and let E_i be the dual modules.*

1. *The map $E_1 \otimes E_2 \to E_1 * E_2$ of page 63 induces a well-defined map $(E_1 \otimes E_2)/\operatorname{Im}(\sigma_1 \otimes \sigma_2 - \mathrm{id}) \to E_1 * E_2$, and this map is bijective iff $2R + b_1R + b_2R = R$.*
2. *If $2R + b_1R + b_2R = R$, the map $A(b, c) \to A_1 \otimes A_2$ of Proposition 2.3 is injective with image $(A_1 \otimes A_2)^{\sigma_1 \otimes \sigma_2}$. It has this property after every base change $R \to S$ iff $2R + b_1R + b_2R = R$.*

The page states the corollary (2) with the hypothesis "after every base change". That hypothesis cannot be dropped: over $R = \mathbb{Z}[X]$ with $b_1 = X$ and $b_2 = 0$, the map is an isomorphism onto the invariants over R itself, although $(2, X)$ is a proper ideal. The example is ours. In (1), the kernel is generated by the Koszul relations of the unimodular row $(2, b_2, b_1)$. Geometrically, the condition says that at every point of residue characteristic 2 one of the two algebras is étale, since $A(b_i, c_i)$ is étale there iff b_i is a unit.

Pages 65–79 build the product of extensions $0 \to L \to E \to M \to 0$ equipped with a χ-splitting, for a fixed $\chi \in R$, by an amalgamated sum and without division. In the split case $E = M \oplus L$, a χ-splitting is given by a linear map $b\colon M \to L$, through $\pi_b(\eta, \xi) = b(\eta) + \chi\xi$.

Theorem 2.5 (Grothendieck, folder 84, pp. 65–79 and 123–131, split case over an affine base)**.** *Let $\chi \in R$, zero included.*

1. *Split extensions $M \oplus L$ with the χ-splitting of $b\colon M \to L$, and morphisms (f, φ, w) satisfying $\varphi b = b' f + \chi w$, form a symmetric monoidal category for the product $b_1 \otimes b_2$. A morphism is invertible as soon as f and φ are.*
2. *For $M = R$, the object (R, L, b) is invertible for the product iff L is an invertible module and $L = Rb + \chi L$ (margin of p. 79).*
3. (After pp. 125, 127 and 169.) *If $\gamma\chi = 2$, the transvection $T_{\gamma b}$ identifies the "pushed" product, with χ-splitting π_b, with the "pulled" product, with π_{-b}. For the unit objects, a map inducing the identity on L and on M and carrying one splitting to the other exists iff some γ satisfies $\gamma\chi = 2$; over $\mathbb{Z}$ with $\chi = 4$ there is none.*

On an affine base every extension of R by L splits, and the amalgamated product of page 68 agrees with the product $b_1 \otimes b_2$; the non-split case over a scheme is not proved here.

3. The line of twists and its degenerate locus

Folder 83 uses the opposite sign: $H = R[u]/(u^2 - bu + c)$ and $H' = R[v]/(v^2 - b'v + c')$, so that $\operatorname{Tr} u = b$ and $N(u) = c$. Pages 92–98 construct, for $(b, c, b', c', t) \in R^5$, the algebra $M = M(b, c; b', c'; t)$, free with basis $1, u, v, uv$, with $u^2 = bu - c$, $v^2 = b'v - c'$, $\operatorname{Tr}(uv) = t$ and the multiplication table of page 92. Pages 126–136 recover it as an algebra structure on $H \otimes H'$. Put

$$Q(t) = t^2 - bb't + b^2c' + b'^2c - 4cc'.$$

Theorem 3.1 (after folder 83, pp. 81–84, 92–98 and 130–143)**.**

1. *For every t, M is an associative unital R-algebra containing H and H'. Every element satisfies $x^2 - \operatorname{Tr}(x)x + \det(x) = 0$,* det *is multiplicative, and $\operatorname{Tr}(uv) = t$. Conversely, let A be an R-algebra whose elements satisfy $x^2 - T(x)x + N(x) = 0$ for a linear form T and a function N, and let $u, v \in A$ have $T(u) = b$, $N(u) = c$, $T(v) = b'$, $N(v) = c'$. Then $x \mapsto x_0 + x_1u + x_2v + x_3uv$ is multiplicative from $M(b, c; b', c'; t)$ to A for some t, unique if this map is injective. Distinct values of t give distinct multiplications, so the structures form the affine line of t.*

(2) (Grothendieck, p. 98.) *M is an Azumaya algebra iff $Q(t)$ is a unit. This holds over every commutative ring, characteristic 2 included, and without hypothesis on $\delta = b^2 - 4c$ and $\delta' = b'^2 - 4c'$. Equivalently, the form* det *on $B = R1 + Ru + Rv$ has invertible half-discriminant, which equals $-Q(t)$.*

(3) The discriminant of the trace form of M in the basis $(1, u, v, uv)$ is $-Q(t)^2$. The polynomial Q has discriminant $\delta\delta'$ and is invariant under $t \mapsto bb' - t$, and the degenerate locus $\widetilde{S} = \operatorname{Spec} R[t]/(Q)$ is free of rank 2 over R.

(4) In $H \otimes H'$, the element $u * v = b'u + bv - 2u \otimes v$ satisfies $Q(u * v) = 0$, which gives a morphism $R[t]/(Q) \to H \otimes H'$. The canonical involution of either factor sends $u * v$ to $bb' - u * v$. For characters φ, φ' of H, H' with values in a commutative R-algebra K, the element $b'\varphi(u) + b\varphi'(v) - 2\varphi(u)\varphi'(v)$ is a root of Q in K.

Sketch. (1) Each axiom is a polynomial identity in the coordinates; for the converse, Hamilton–Cayley applied to $u + v$ gives $vu = (t - bb') + b'u + bv - uv$ with $t = bb' + c + c' - N(u + v)$, and the rest of the table follows by associativity. (2) M is free, so it is Azumaya iff the sandwich map $M \otimes M^{\mathrm{op}} \to \operatorname{End}_R M$ is bijective, that is, iff its 16×16 determinant D is a unit. Composing with the trace pairing of $\operatorname{End}_R M$ gives the Kronecker square of the trace form of M, so $\pm D^2 = (-Q^2)^8 = Q^{16}$. The argument uses only the determinant of the trace form, not its non-degeneracy, so it holds in characteristic 2. (4) In $H \otimes H'$, with $U = u \otimes 1$ and $V = 1 \otimes v$,

$$Q(b'U + bV - 2UV) = (2V - b')^2(U^2 - bU + c) + (b^2 - 4c)(V^2 - b'V + c'),$$

and the right-hand side vanishes. □

Two misprints on the pages are corrected here. Page 138 writes the constant term of Q as $b^2b' + b'^2b - 4cc'$, and page 143 writes $\det(u * v) = b^2c + b'^2c' - 4cc'$; both should read $b^2c' + b'^2c - 4cc'$. For $b = c = b' = c' = 0$, $M(0, 0; 0, 0; t)$ is Azumaya iff t is a unit, as page 97 says.

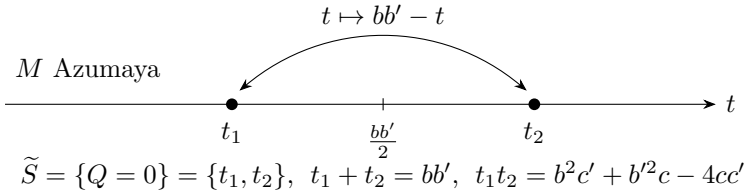


FIGURE 1. The line of twists over a field, when Q has two roots. Away from t_1 and t_2 the algebra $M(b, c; b', c'; t)$ is Azumaya. The canonical involutions exchange the two degenerate points, and the two roots coincide iff $\delta\delta' = 0$.

Point (4) links this section with Section 2. In the notation of folder 84, $H = A(-b, c)$ and $H' = A(-b', c')$. The generator of Proposition 2.3 is then $\mathcal{U} = 2UV - bV - b'U = -(u * v)$, and the product is $A(bb', b^2c' + b'^2c - 4cc')$, with equation $P(T) = T^2 + bb'T + b^2c' + b'^2c - 4cc'$. Since $Q(t) = P(-t)$, the map $t \mapsto -\mathcal{U}$ identifies $R[t]/(Q)$ with the product of H and H', and the morphism of (4) with the canonical morphism of Proposition 2.3. In Voight's convention, Q is exactly the polynomial of his Construction 3.12. So the non-Azumaya locus of the line of twists is the spectrum of the product $H * H'$, as pages 137–143 assert. Pages 137–143 also identify $\widetilde{S}$ with the quotient of $X \times_S X'$ by $\sigma \times \sigma'$. We prove only the algebraic core of that statement: by Theorem 2.4, the invariants of $\sigma \otimes \sigma'$ in $H \otimes H'$ are $R[u * v]$ when 2, b, b' generate the unit ideal.

4. Rotations in forms of PGL_2

Folder 85 realises polygons and polyhedra in a form of PGL_2 over a base by means of a single invariant. For $g \in M_2(R)$ with $\det g$ invertible, put

$$A(g) = \frac{(\operatorname{Tr} g)^2}{\det g} - 2.$$

It depends only on the image of g in $\mathrm{PGL}_2(R)$, and it equals $\zeta + \zeta^{-1}$ when the eigenvalues of g have ratio ζ. Call g *nowhere scalar* if its image in $M_2(R/\mathfrak{m})$ is not scalar for every maximal ideal $\mathfrak{m}$; equivalently, the off-diagonal entries and the difference of the diagonal entries generate the unit ideal. For $\nu \geq 2$, say that g has *order ν at every point* if g^ν is scalar and g^m is nowhere scalar for every divisor m of ν with $1 \leq m < \nu$. For ν prime and g nowhere scalar, this just says that g^ν is scalar.

Theorem 4.1 (after folder 85, pp. 6–7, 29–31 and 40)**.** *Let $g \in M_2(R)$ be nowhere scalar, with trace t, invertible determinant d, and $\alpha = A(g)$.*

(1) With $x_0 = 0$, $x_1 = 1$, $x_{n+2} = tx_{n+1} - dx_n$, one has $g^{n+1} = x_{n+1}g - dx_n$, and g^{n+1} is scalar iff $x_{n+1} = 0$. In particular g^2 is scalar iff $t = 0$, that is, $\alpha = -2$.

(2) g^3 is scalar iff $\alpha = -1$. The element g has order 4 at every point iff $\alpha = 0$ and 2 is invertible, and order 6 at every point iff $\alpha = 1$ and 6 is invertible.

(3) g^5 is scalar iff $\alpha^2 + \alpha - 1 = 0$. Moreover $A(g^2) = A(g)^2 - 2$, so that $A(g^2) = -1 - A(g)$ in that case (the corollary of page 40).

The implication "$\alpha^2 + \alpha - 1 = 0 \Rightarrow g^5$ scalar" holds for every g with invertible determinant.

Proof. (1) is Cayley–Hamilton, $g^2 = tg - d$, together with the fact that $xg + y$ is scalar iff $x = 0$ when g is nowhere scalar. Since $t^2 = (\alpha + 2)d$, induction gives $x_{2k+1} = d^k P_{2k+1}(\alpha)$ and $x_{2k+2} = t\,d^k P_{2k+2}(\alpha)$, where $P_0 = 0$, $P_1 = 1$, $P_{n+2} = P_{n+1} - P_n$ for n even and $P_{n+2} = (\alpha+2)P_{n+1} - P_n$ for n odd. Thus $P_3 = \alpha + 1$, $P_4 = \alpha$, $P_5 = \alpha^2 + \alpha - 1$ and $P_6 = \alpha^2 - 1$, which gives (2) and (3). For order 4, g^2 is nowhere scalar iff $x_2 = t$ is invertible, and then $\alpha = 0$ gives $t^2 = 2d$, so 2 is invertible. For order 6, t and $x_3 = d(\alpha + 1)$ must be invertible, so $\alpha = 1$, and then $t^2 = 3d$ and $\alpha + 1 = 2$ make 6 invertible. Finally $A(g^2) = (t^2 - 2d)^2/d^2 - 2 = A(g)^2 - 2$. □

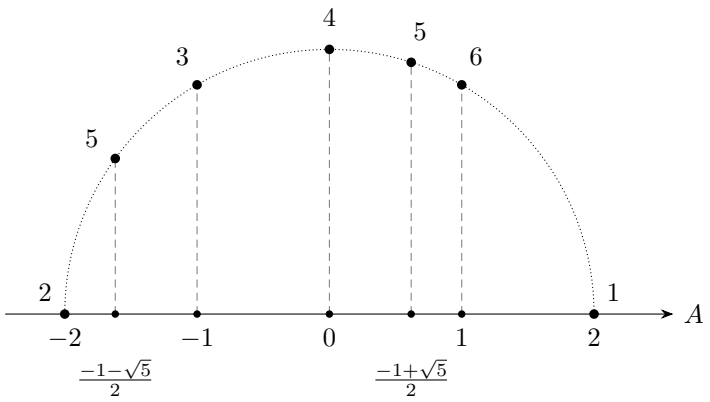


FIGURE 2. Reading the order on $A(g)$. Over $\mathbb{C}$, a rotation by the angle θ has $A = 2\cos\theta$. The points of the upper half circle are labelled by the order of the rotation, and their projections are the values of A: -2 for involutions, -1, 0 and 1 for orders 3, 4 and 6, and the two roots of $A^2 + A - 1$ for order 5. Theorem 4.1 proves the same reading over any ring, with the conditions on 2 and 6 for orders 4 and 6.

Example 4.2 (folder 85, p. 40). Over $\mathbb{Z}[\zeta_5]$, the matrices $s = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $p = \left(\begin{smallmatrix} \zeta & 1 \\ 0 & \zeta^4 \end{smallmatrix}\right)$ satisfy $s^2 = -1$, $(sp)^3 = -1$ and p^5 scalar, and $A(p) = \zeta^2 + \zeta^3$ is a root of $T^2 + T - 1$. The matrices

s, p and sp are nowhere scalar. Since $\langle s, p \mid s^2, (sp)^3, p^5\rangle$ is a presentation of $\mathfrak{A}_5$, they define a homomorphism $\mathfrak{A}_5 \to \mathrm{PGL}_2(\mathbb{Z}[\zeta_5])$. It is injective on every fibre, because its image there is a non-trivial quotient of the simple group $\mathfrak{A}_5$. This last step is not formalised. In characteristic 5, $\zeta = 1$ and p is unipotent, with $A(p) = 2$, a double root of $T^2 + T - 1 = (T - 2)^2$.

Page 40 states the following.

Statement 4.3 (Grothendieck, folder 85, p. 40)**.** For a form G of PGL_2 over a scheme S and a monomorphism $\varphi\colon \mathfrak{A}_5 \to G$, the element $\alpha(\varphi) = A(\varphi(\pi))$, π a 5-cycle, satisfies $\alpha^2 + \alpha - 1 = 0$. The category of pairs (G, φ) is rigid, and its functor of isomorphism classes is represented by $\operatorname{Spec} \mathbb{Z}[T]/(T^2 + T - 1)$.

Theorem 4.1(3) proves the first assertion over every base, for matrices. We do not prove rigidity or representability. Together they would say that, over any base, including residue characteristics 2, 3 and 5, a pair (G, φ) is determined up to unique isomorphism by the root α. Forms of PGL_2 cannot be avoided here. By Beauville's Proposition 1.1 [2], for a field K of characteristic prime to 60, $\mathrm{PGL}_2(K)$ contains $\mathfrak{A}_5$ iff -1 is a sum of two squares in K and 5 is a square. So the split group over $\mathbb{R}$, or over $\mathbb{Q}(\sqrt{5})$, contains no $\mathfrak{A}_5$.

The group G_2. Pages 4–7 treat homomorphisms into $\mathrm{PGL}_2(R)$ of the group $G_2 = \langle \tau_0, \tau_1, \tau_2 \mid \tau_i^2, (\tau_0\tau_2)^2\rangle$. The notation is the page's; G_2 is the Coxeter group with string diagram and unspecified orders of $\tau_0\tau_1$ and $\tau_1\tau_2$, not the exceptional group. Such a homomorphism is given by matrices τ_i with invertible determinants and with τ_i^2 and $(\tau_0\tau_2)^2$ scalar. Put $\rho_0 = \tau_0\tau_1$, $\rho_1 = \tau_1\tau_2$, $\alpha_i = A(\rho_i)$, and $\Delta = (\alpha_0 + 2)(\alpha_1 + 2)(\alpha_0 + \alpha_1)$. Over a field, condition $(*)$ of page 4 says that ρ_0^2 and ρ_1^2 are not scalar and that $\rho_0\rho_1$ is not a scalar multiple of $\rho_1\rho_0$. We say that $(*)$ holds at every point if it holds for the images of the τ_i in $M_2(R/\mathfrak{m})$ for every maximal ideal $\mathfrak{m}$.

Theorem 4.4 (after folder 85, pp. 4–7)**.**

(1) $(*)$ *holds at every point iff τ_0, τ_1, τ_2 and $\tau_0\tau_2$ are traceless and Δ is invertible. Each factor of Δ corresponds to one condition of* $(*)$.

(2) Rigidity. *Under these conditions, if $g \in \mathrm{GL}_2(R)$ and $\mu_i \in R$ satisfy $g\tau_i = \mu_i\tau_i g$ for $i = 0, 1, 2$, then g is scalar.*

(3) Normal form. *For every (α_0, α_1) with Δ invertible, explicit traceless matrices with these invariants and with $(*)$ at every point exist over $R[T]/(T^2 - \alpha_0 T + 1)$, which is free of rank 2 over R. Over a field containing a root of $T^2 - \alpha_0 T + 1$, two triples with the same invariants are conjugate in* PGL_2.

Sketch. For traceless τ_i with $\mathrm{Tr}(\tau_0\tau_2) = 0$, write $t_{01} = \mathrm{Tr}\,\rho_0$, $t_{12} = \mathrm{Tr}\,\rho_1$, $T = \mathrm{Tr}(\tau_0\tau_1\tau_2)$ and $d_i = \det \tau_i$. Then $\alpha_0 + 2 = t_{01}^2/(d_0d_1)$, $\alpha_1 + 2 = t_{12}^2/(d_1d_2)$, $\rho_0\rho_1 - \rho_1\rho_0 = -T\tau_1$ and $\alpha_0 + \alpha_1 = -T^2/(d_0d_1d_2)$, so Δ is invertible iff t_{01}, t_{12} and T are. Conversely, an involution τ that is nowhere scalar has $\tau^2 = (\mathrm{Tr}\,\tau)\tau - \det\tau$ scalar, which forces $\mathrm{Tr}\,\tau = 0$. For (2), the traces give $\mu_0\mu_1 = \mu_1\mu_2 = 1$, so g commutes with ρ_0 and ρ_1; then $g = p + q\rho_0$, and $qT\tau_1 = 0$ forces $q = 0$. □

The moduli statement that follows is the edition's reformulation of pages 4–7 and is not proved here: homomorphisms of G_2 into forms of PGL_2 satisfying $(*)$ in every fibre would be classified by the complement of the three lines $\Delta = 0$ in $\mathbb{A}^2_{\mathbb{Z}}$.

Octahedral configurations. Folder 64 considers a faithful action of the Klein group on $\mathbb{P}^1$, locally $z \mapsto -z$, $z \mapsto 1/z$, whose axes form the octahedral configuration $\Delta = \{0, \infty, \pm 1, \pm i\}$. To identify $\mathrm{Aut}(\mathbb{P}^1, \Delta)$ with the rotation group of the octahedron, page 48 computes the fixer of the two non-opposite vertices ∞ and -1. Its elements are the maps $z \mapsto az + (a - 1)$.

Theorem 4.5 (Grothendieck, folder 64, p. 48)**.** *Let K be a field with $2 \neq 0$ and $i^2 = -1$. If $z \mapsto az + (a - 1)$ maps $\{0, 1, i, -i\}$ into itself, then $5(a - 1) = 0$. In particular, if $5 \neq 0$ in K, the fixer of ∞ and -1 in $\mathrm{Aut}(\mathbb{P}^1_K, \Delta)$ is trivial.*

Proof. The four points are distinct because $2 \neq 0$. If $a = 0$, the map sends 0 to -1, which is not one of them. Otherwise the map is injective, so it permutes the four points and preserves their sum, which is 1. The sum of the images is $a + 4(a-1) = 5a - 4$, so $5a - 4 = 1$. □

The margin restricts the computation to "car. $\neq 3$ et 5". The proof shows that characteristic 3 needs no special case. In characteristic 5 the identification fails. There $i = \pm 2$, and $z \mapsto 2z+1$ permutes $\{0, 1, i, -i\}$. Over $\mathbb{F}_5$ the configuration Δ is all of $\mathbb{P}^1(\mathbb{F}_5)$, so every one of the 120 elements of $\mathrm{PGL}_2(\mathbb{F}_5)$ preserves it, and the fixer of ∞ and -1 consists of the identity and the maps $z \mapsto az + (a-1)$ for $a = 2, 3, 4$ (Figure 3).

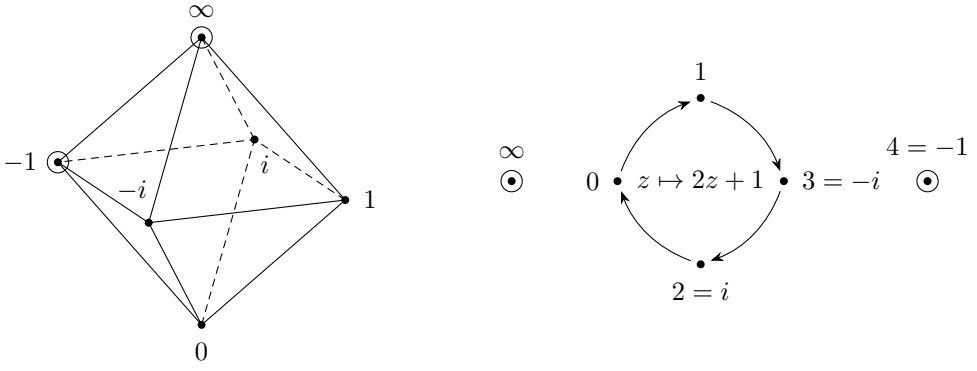


FIGURE 3. (a) The octahedral configuration; opposite vertices are 0 and ∞, 1 and -1, i and $-i$. The fixer of the circled vertices ∞ and -1 permutes the other four, and Theorem 4.5 shows it is trivial when $5 \neq 0$. (b) Over $\mathbb{F}_5$, with $i = 2$, the six vertices are the six points of $\mathbb{P}^1(\mathbb{F}_5)$, and $z \mapsto 2z+1$ fixes ∞ and -1 and cycles the other four.

Pages 44–52 state that, over a base whose residue characteristics are $\neq 2$, three sets correspond bijectively: the subgroup schemes of $\mathrm{Aut}_S(X)$ locally isomorphic to $(\mathbb{Z}/2)^2$, those locally isomorphic to $\mathfrak{S}_4$, and the regular octahedral configurations. Where the pages identify $\mathrm{Aut}_S(X, \Delta)$ with the octahedral rotation group, the example above shows that 5 must be invertible. Over a field, Beauville's Proposition 1.1 [2] says that $\mathrm{PGL}_2(K)$ contains $\mathfrak{S}_4$ iff -1 is a sum of two squares, for K of characteristic prime to 6; in characteristic 5 this subgroup exists, but it is not the full stabiliser of Δ. Pages 74–91 describe the moduli of elliptic curves with a level-2 structure and a "Legendre pinning" over $\mathbb{Z}[1/2]$ by $U_{0,3} \times \mu_4^*$. Antieau and Meier prove a related statement without the pinning: the parameter space of Legendre curves $\mathbb{A}^1_{\mathbb{Z}[1/2]} - \{0, 1\}$ is a C_2-torsor over the stack $\mathcal{M}(2)$ of elliptic curves with full level-2 structure [1, Definition 4.4 and Proposition 4.5, arXiv numbering]. Neither of the folder's relative statements is proved here.

5. The Schwarzian with divided derivatives

Folder 93 studies the Schwarzian derivative and projective connections on a smooth curve over a base. For two uniformisers t, s, the classical Schwarzian difference is $G(t, s) = (-2s'''/s' + 3(s''/s')^2)\, dt^2$. Page 72 replaces the derivatives by the divided operators $D^{(i)} = \frac{1}{i!}\frac{d^i}{dt^i}$ of Hasse and Schmidt [3], observes that the coefficients 2 and 3 of G prevent this, and takes $G_0 = G/12$:

$$G_0(t, s) = -\frac{D^{(3)}s}{D^{(1)}s} + \Big(\frac{D^{(2)}s}{D^{(1)}s}\Big)^2.$$

Put $N(s) = (D^{(2)}s)^2 - D^{(1)}s \cdot D^{(3)}s$, so that $G_0(s) = N(s)/(D^{(1)}s)^2$. We work in the local model $R[X]$, with X in the role of t, and in its quotients.

Theorem 5.1 (Grothendieck, folder 93, pp. 71–73 and 115–123)**.**

(1) $G = 12\,G_0$, *and* G_0 *is defined over every base, characteristics* 2 *and* 3 *included.*
(2) Cocycle. *For polynomials* s, u, *one has* $N(s \circ u) = (N(s) \circ u)(D^{(1)}u)^4 + \big((D^{(1)}s) \circ u\big)^2 N(u)$. *If* $D^{(1)}s$ *and* $D^{(1)}u$ *are invertible,* $G_0(s \circ u) = (G_0(s) \circ u)(D^{(1)}u)^2 + G_0(u)$.
(3) Homographies. *If* $a, b, c, d \in R$, $cX + d$ *is invertible and* $s(cX + d) = aX + b$, *then* $G_0(s) = 0$.
(4) Killed functions *(margin of p. 121). In characteristic* p, *a polynomial is killed by* $D^{(1)}$, $D^{(2)}$ *and* $D^{(3)}$ *iff it is a polynomial in* X^p *for* $p > 3$, *in* X^4 *for* $p = 2$, *and in* X^9 *for* $p = 3$.
(5) Rigidity. *Let* A *be a local ring of characteristic* p, *let* $e = 1$ *if* $p > 3$ *and* $e = 2$ *if* $p = 2, 3$, *and let* $\alpha \in A$. *If* $s \in A[T]/(T^{p^e} - \alpha)$ *has* $D^{(1)}s$ *invertible and* $G_0(t, s) = 0$, *then* $s = (at + b)/(ct + d)$ *with* $a, b, c, d \in A$ *and* $ct + d$ *invertible. The same holds for any* A *of characteristic* p *with* $\mathrm{Pic}(A) = 0$.

Sketch. (1)–(3) are identities of the divided-power calculus. For (3), Leibniz applied to $sq = p$, with $q = cX + d$ and $p = aX + b$, gives $q^2 N(s) = 0$. For (5) with $\alpha = 0$, take $d = 1$, $c = -s_2/s_1$, $b = s_0$, $a = s_1 + cs_0$ and $r = s\,(cX + 1)$. If X^{j+3} divides $r - (aX + b)$, the coefficient of X^j in $(cX + 1)N(s)$ is $-\binom{j+3}{3} s_1 r_{j+3}$. Induction on j then gives $X^n \mid s(cX + 1) - (aX + b)$ as long as $\binom{k}{3}$ is invertible for $3 \le k < n$. This holds for $n = p$ ($p > 3$), $n = 4$ ($p = 2$, where only $\binom{3}{3} = 1$ occurs) and $n = 9$ ($p = 3$, where $\binom{k}{3} = 1, 4, 10, 20, 35, 56$ for $k = 3, \dots, 8$). For general α, let $q = p^e$ and $B = A[T]/(T^q - \alpha)$, where $T^q - \alpha = (T - \beta)^q$ with β the class of T. The Taylor map $\theta(f) = f(\beta + Y)$ reduces the question to the case $\alpha = 0$ over B. This gives the osculating homography at β, with coefficients in B. Its coefficient vector v satisfies $(cY + 1)\,\theta(v) \equiv v \bmod Y^q$, so for ℓ in $Ac + A(1 - cT)$ the vector $\ell^{q-1}v$ is killed by the $D^{(k)}$, $0 < k < q$, and lies in A^4. Over a local ring, c or $1 - cT$ is invertible, since $(1 - cT)^q = 1 - c^q\alpha$. When $\mathrm{Pic}(A) = 0$, the denominators of the homographies equal to s form an invertible A-submodule of B, which is then principal with a unit generator. □

We correct the pages in two places. First, the determinant cannot be normalised to $ad - bc = 1$, as page 121 does. In $\mathbb{F}_5[T]/(T^5)$, $s = 2t$ has $N(s) = 0$ and $D^{(1)}s = 2$ invertible, but 2 is not a square in $\mathbb{F}_5$, so s is not a homography with coefficients in $\mathbb{F}_5$ and $ad - bc = 1$. The correct statement is in $\mathrm{PGL}_2(A)$. Second, the page passes from $\alpha \neq 0$ to $\alpha = 0$ without argument ("OPS"); the descent above supplies the step over a local ring, and more generally when $\mathrm{Pic}(A) = 0$.

The level p^2 at $p = 2, 3$ comes from the higher divided derivatives in G_0. Since $D^{(k)}(X^p) = \binom{p}{k} X^{p-k}$, the p-th power X^p is killed by $D^{(1)}$, $D^{(2)}$, $D^{(3)}$ iff $p > 3$. For $p = 2$, $D^{(2)}(X^2) = 1$, and the translation $t \mapsto t + t^2$ has $G_0 = dt^2$; for $p = 3$, $D^{(3)}(X^3) = 1$, and $t \mapsto t + t^3$ has $G_0 = -dt^2$. So for $p = 2, 3$ the group of homographies with coefficients in $\mathcal{O}^{(p)}$ does not leave G_0 invariant. The primes involved are exactly those dividing the factor $12 = G/G_0$.

6. Verification

The proofs above are formalised in Lean 4 with Mathlib, in the files `Folder84.lean`, `Folder83.lean`, `Folder85.lean`, `Folder64.lean` and `Folder93.lean` of the repository Commutator-IO/grothendieck-project (directory `lean/Grothendieck`). They use only Lean's standard axioms. The main theorem names are as follows.
Section 2. Theorem 2.1: `tripletCond_iff_forall`, `isCoverHom_triplet`, `triplet_ess_surj`, `triplet_full_faithful`, `triplet_classification`, `cover_iso_iff_triplet_iso`, `tripletCond_baseChange`, `isCoverHom_baseChange`, `tripletHom_baseChange`, `int_disc_iff`, `int_qalg_iso_iff`, `two_regular_needed`, and for the reconstruction `recon_one`, `recon_mul`, `recon_norm`, `div_norm`, `div_prod_mem`. Proposition 2.3: `identity_page64`, `disc_mul`, `tensor_root`, `productToTensor_root`. Theorem 2.4: `coinvariants_bijective_iff`, `invariants_of_span_eq_top`, `invariants_universally_iff`, `invariants_not_iff`. Theorem 2.5: the instances `MonoidalCategory` and `SymmetricCategory` on `ChiExt`, `ChiExt.isIso_of`,

`ChiExt.invertible_iff_span`, `push_pull`, `push_pull_unit_iff`, `no_gamma_four`, and for the amalgamated product `isPushout`, `prodSplitting_b`, `biaffine_universal`.
Section 3. `mulM_assoc`, `hamilton_cayley`, `nrdM_mulM`, `realise_mulM`, `realise_unique`, `mulM_inj`, `isAzumaya_iff`, `isAzumaya_iff_deltaPB`, `isAzumaya_nilpotent_iff`, `deltaPB_eq`, `det_traceForm`, `disc_Q`, `Q_invol`, `Stilde_rank`, `Q_star_identity`, `star_root`, `conjL_star`, `conjR_star`, `twist_root`.
Section 4. Theorem 4.1: `invA_smul`, `pow_succ_eq`, `isScalar_pow_iff`, `powCoeff_eq_chebP`, `isScalar_pow_three_iff`, `order_four_iff`, `order_six_iff`, `sq_add_sub_of_pow_five`, `pow_five_of_sq_add_sub`, `invA_sq`, `invA_sq_of_root`. The example: `exists_model_adjoinRoot`. Theorem 4.4: `discr_isUnit_of_star`, `star_of_discr_isUnit`, `rigid`, `exists_normalForm_adjoinRoot`, `conj_of_same_invariants`. Theorem 4.5 and characteristic 5: `fixer_five`, `fixer_trivial`, `fixer_nontrivial_char5`, `octahedral_eq_univ`, `card_pgl`, `card_fixer`.
Section 5. `Ncl_eq`, `G_eq`, `N_comp`, `G0_comp`, `G0_homography_const`, `killed_iff_of_gt`, `killed_iff_two`, `killed_iff_three`, `char_two`, `char_three`, `rigidity_of_gt`, `rigidity_four`, `rigidity_nine`, `rigidity_alpha_local`, `rigidity_alpha_of_gt`, `rigidity_alpha_four`, `rigidity_alpha_nine`, `rigidity_alpha_pic`, `det_one_counterexample`. In Lean, $G_0(t,s)=0$ in B is the hypothesis $N(s)=0$ in B, which is equivalent when $D^{(1)}s$ is invertible.
Not proved here: the global form of Theorem 2.1 on schemes; the intrinsic product of non-split χ-extensions over a scheme; the identification of $\widetilde{S}$ with the quotient of $X\times_S X'$; the injectivity on fibres of the homomorphism of the example; rigidity and representability in Statement 4.3 and in the G_2 moduli; the relative octahedral correspondence and the Legendre pinnings of folder 64.

Statement on the use of artificial intelligence

The transcriptions of folders 64, 83, 84, 85 and 93, the modernised readings, the proofs, the Lean formalisation and a first draft of this text were produced with language models (Claude Opus 5 and Opus 5.5, Anthropic), under the author's direction. The author checked the statements, the proofs and the references, and takes full responsibility for the content. Lean checks the formal proofs mechanically.

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